

Program 9 : Organize Kubernetes pods using labels and namespaces

Project Structure

```
├── namespaces.yaml
└── pods.yaml
```

Step 1 : Create namespaces.yml file

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ cat namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$
```

Step 2 : Create a pods.yml file

```
name: prod1rv24mc048_Jayesh@Jayesh:~/Desktop,
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$
```

Step 3 : Apply Namespace and pods

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k apply -f namespaces.yaml
namespace/dev created
namespace/prod created
```

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 4 : Verify pods in each namespace

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k get pods -n dev
NAME           READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           20s
frontend-pod   1/1     Running   0           20s
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k get pods -n prod
NAME           READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           27s
frontend-pod   1/1     Running   0           27s
```

Step 5 : Filter pods by labels

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k get pods -n dev -l app=frontend
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           91s
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k get pods -n prod -l app=frontend
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0          104s
```

Step 6 : Clean up process

```
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k delete pods --all
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k delete svc --all
service "kubernetes" deleted
1rv24mc048_Jayesh@Jayesh:~/Desktop/DEVOPS_Lab/lab9$ k delete deployments --all
```